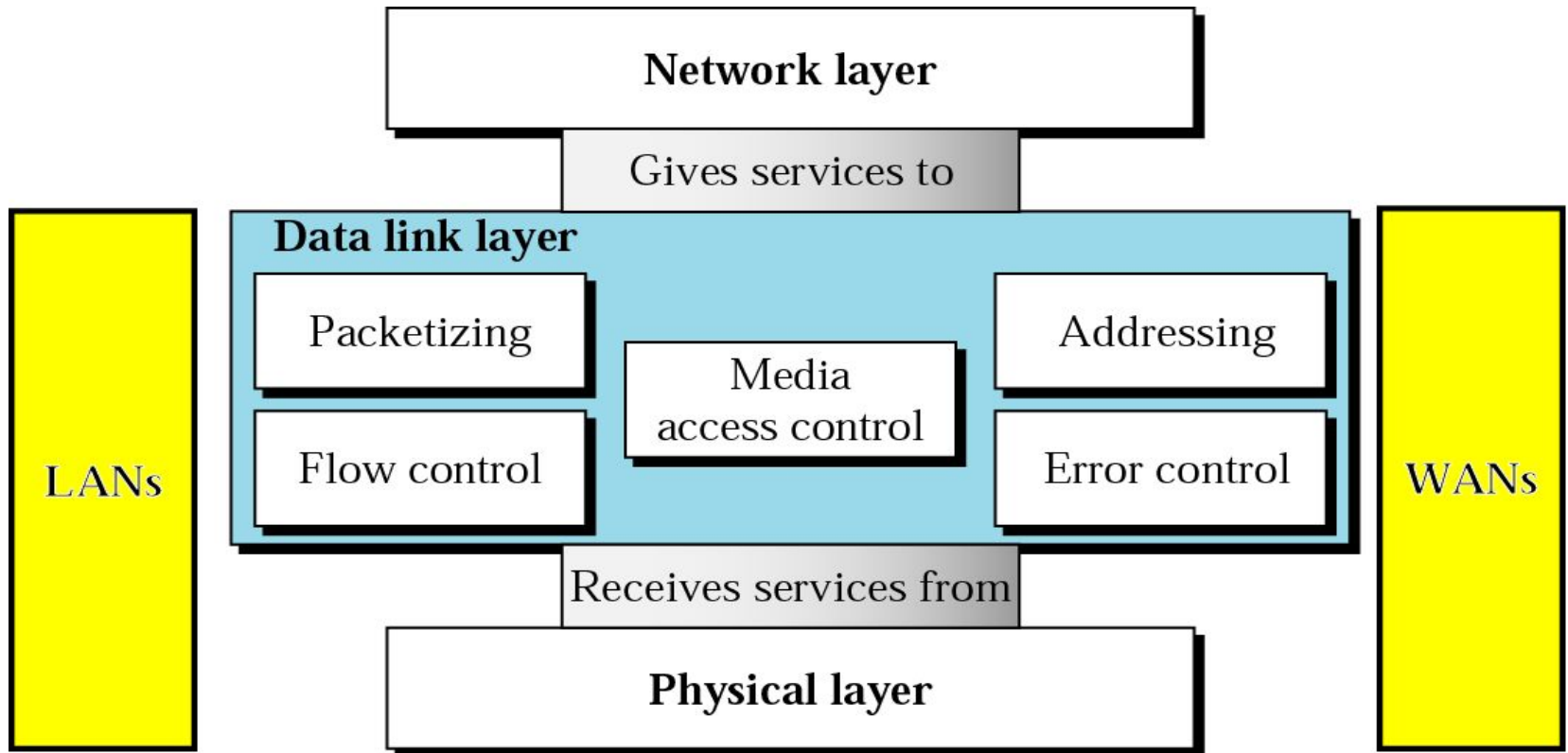


Computer Communication & Networks

Lecture 20

Datalink Layer: Wireless LANs

Data Link Layer



Data Link Layer Topics to Cover

Error Detection and Correction

Data Link Control and Protocols

Multiple Access

Local Area Networks

Wireless LANs

IEEE 802 Series of LAN Standards

- 802 standards free to download from <http://standards.ieee.org/getieee802>

IEEE 802®: Overview & Architecture

IEEE 802.1™ Bridging & Management

IEEE 802.2™: Logical Link Control

IEEE 802.3™: CSMA/CD Access Method

IEEE 802.4™: Token-Passing Bus Access Method

IEEE 802.5™: Token Ring Access Method

IEEE 802.6™: DQDB Access Method

IEEE 802.7™: Broadband LAN

IEEE 802.10™: Security

IEEE 802.11™: Wireless

IEEE 802.12™: Demand Priority Access

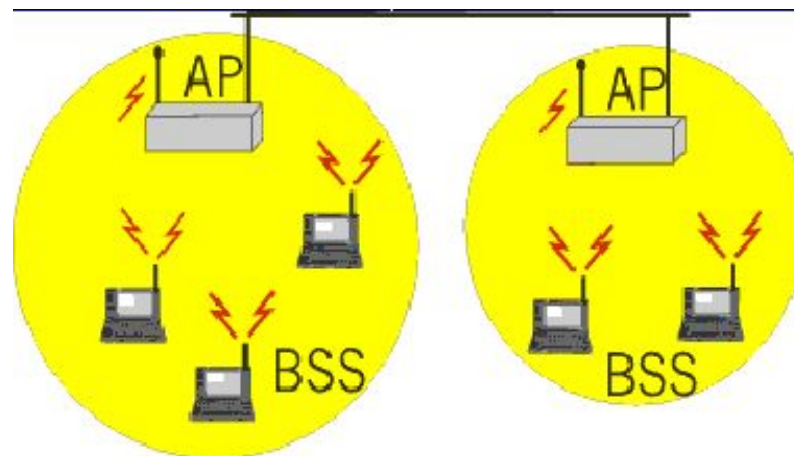
IEEE 802.16™: Broadband Wireless Metropolitan Area Networks



WiMAX →

IEEE 802.11 Wireless LAN - 1

- **Wireless LANs**
- **IEEE 802.11 standard:**
 - **MAC protocol**
 - **unlicensed frequency spectrum: 900Mhz, 2.4Ghz**



IEEE 802.11 Wireless LAN - 2

- **Basic Service Set (BSS)** (a.k.a. “cell”) contains:
 - wireless hosts
 - **access point (AP):** base station
- **BSS's combined to form distribution system (DS)**

Ad Hoc Networks

- **Ad hoc network:** IEEE 802.11 stations can dynamically form network *without* AP
- **Applications:**
 - “laptop” meeting in conference room, car
 - interconnection of “personal” devices
 - battlefield
- **IETF MANET**
(Mobile Ad hoc Networks)
working group



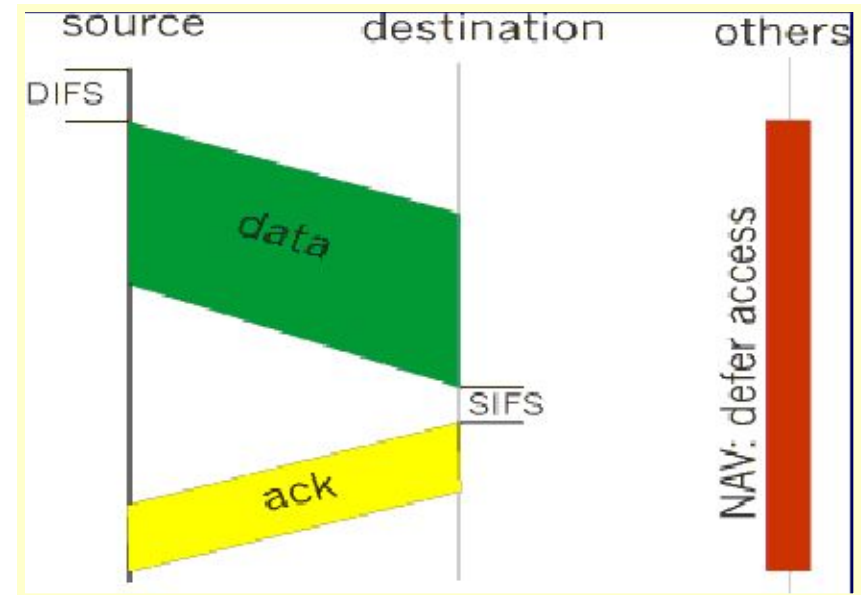
IEEE 802.11 MAC Protocol: CSMA/CA

802.11 CSMA sender:

- if sense channel idle for DIFS sec.
then transmit entire frame (no collision detection)
- if sense channel busy
then binary backoff

802.11 CSMA receiver:

- if received OK
return ACK after SIFS

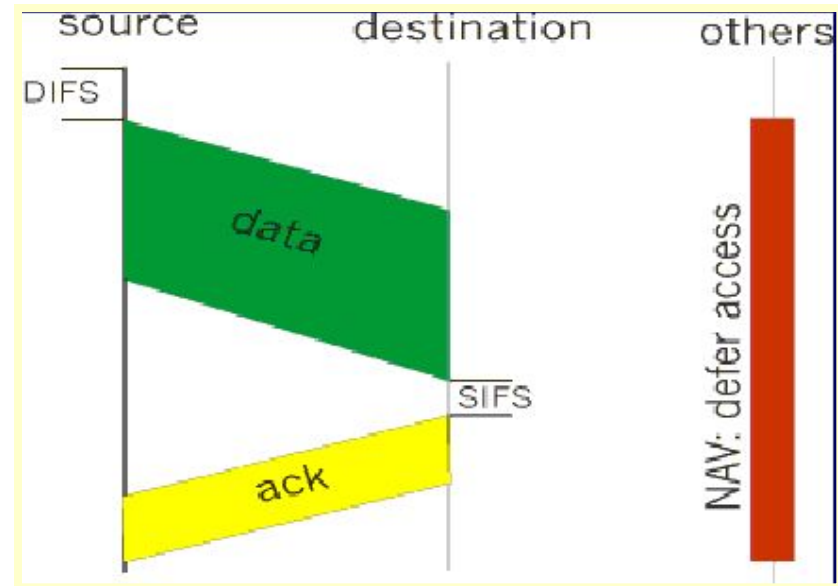


IEEE 802.11 MAC Protocol

802.11 CSMA Protocol:

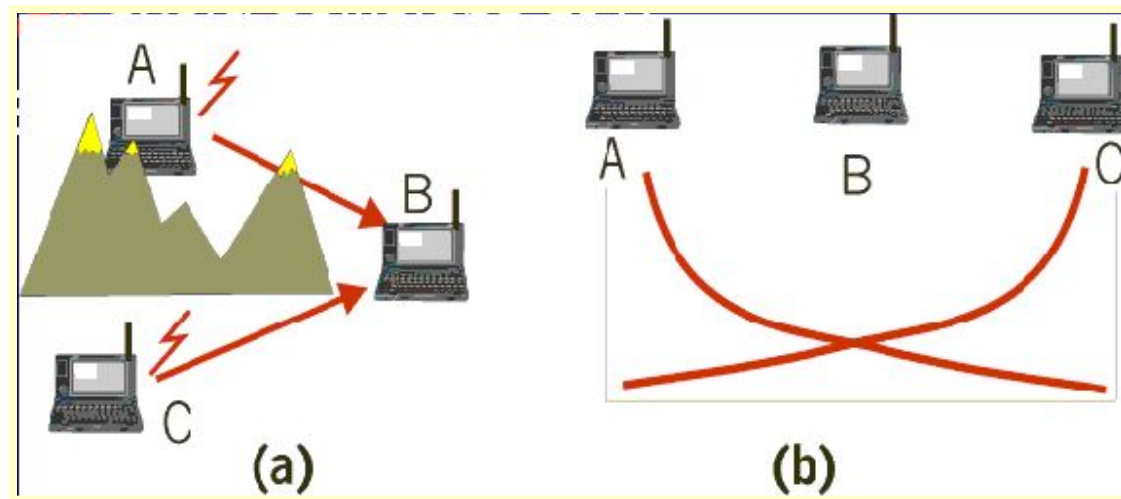
others

- **NAV: Network Allocation Vector**
- **802.11 frame has transmission time field**
- **others (hearing sata) defer access for NAV time units**

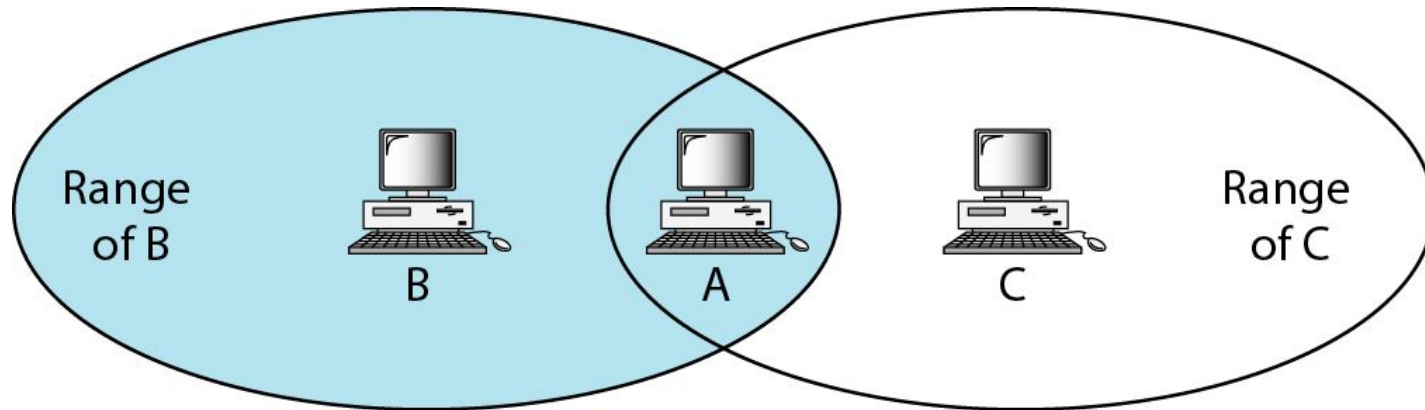


Hidden Terminal effect

- **hidden terminals:** A, C cannot hear each other
 - ❑ obstacles, signal attenuation
 - ❑ collisions at B
- **goal:** avoid collisions at B
- **CSMA/CA:** CSMA with Collision Avoidance



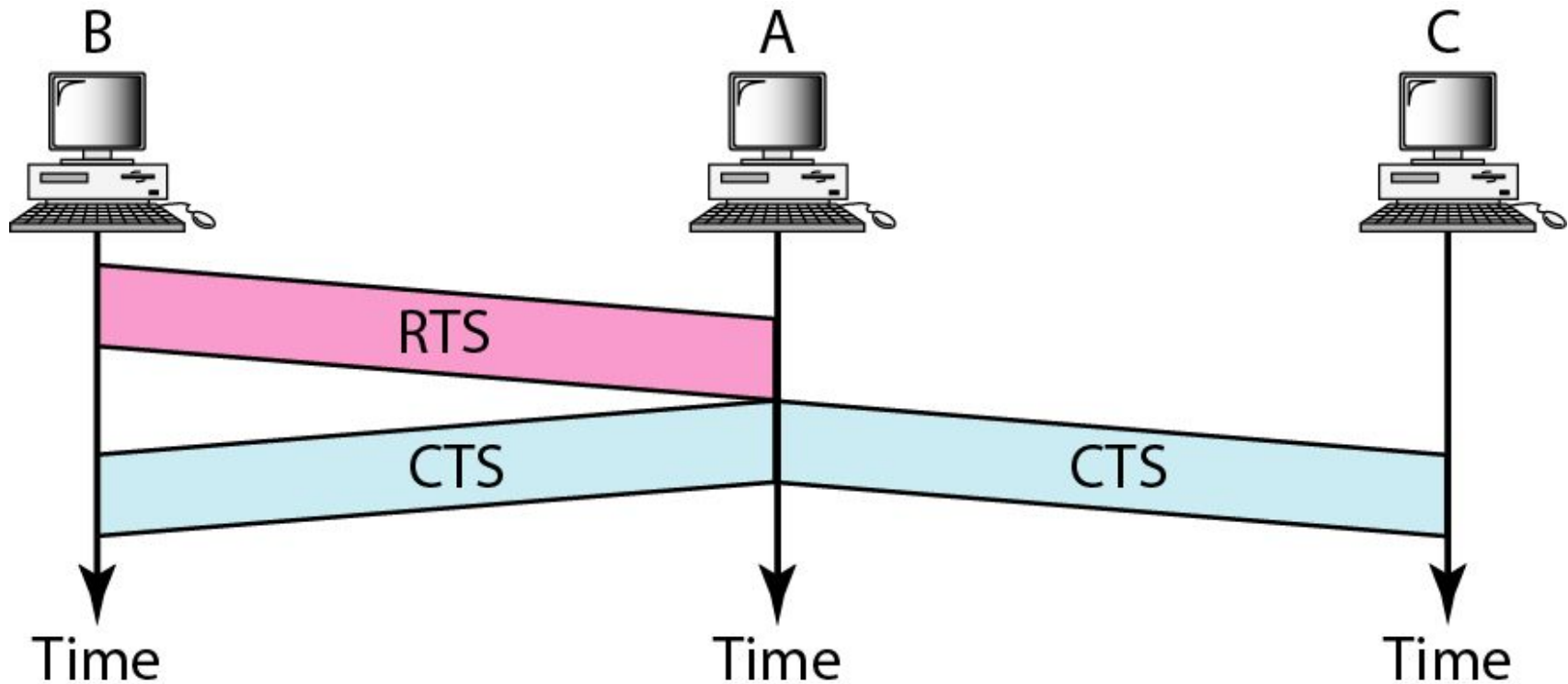
Hidden Station Problem



B and C are hidden from each other with respect to A.

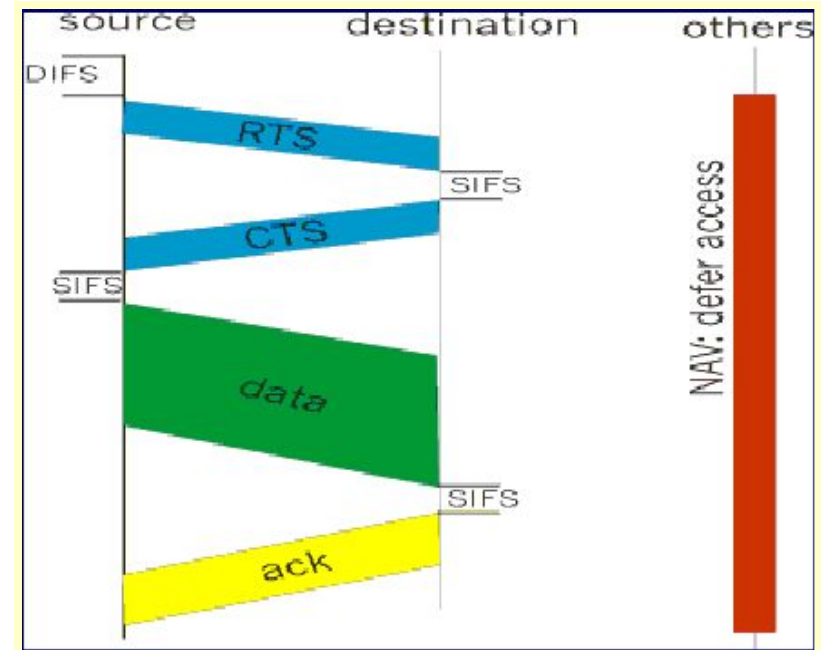
- The CTS frame in CSMA/CA handshake can prevent collision from a hidden station.

Use of Handshaking to Prevent Hidden Station Problem



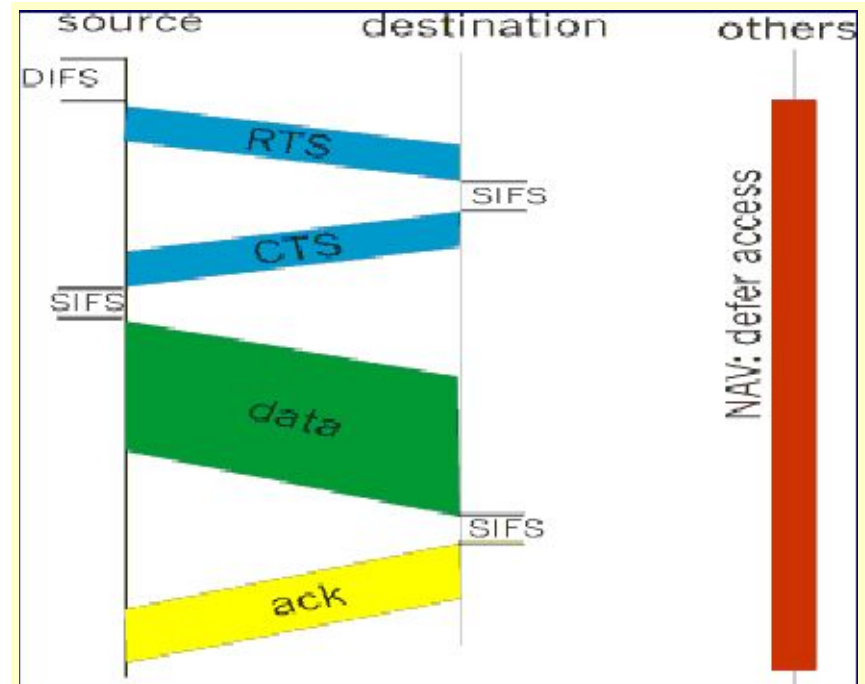
Collision Avoidance: RTS-CTS exchange - 1

- **CSMA/CA: explicit channel reservation**
 - ❑ sender: send short RTS: request to send
 - ❑ receiver: reply with short CTS: clear to send
- **CTS reserves channel for sender, notifying (possibly hidden) stations**
- **Avoid hidden station collisions**

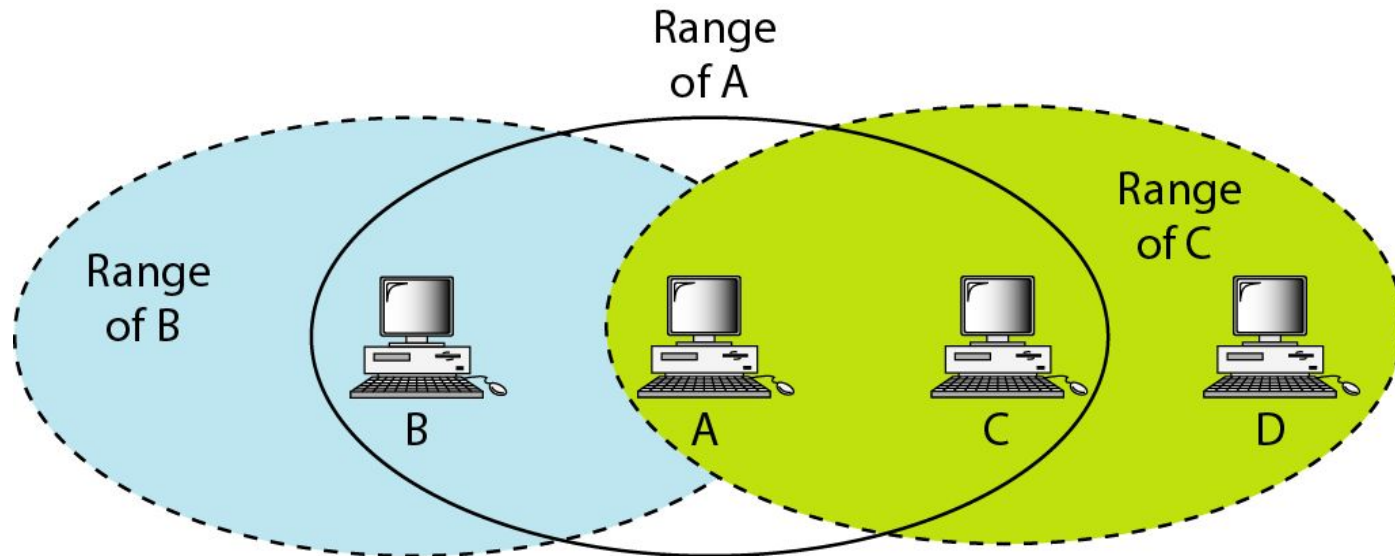


Collision Avoidance: RTS-CTS exchange - 2

- **RTS and CTS short:**
 - ❑ collisions less likely, of shorter duration
 - ❑ end result similar to collision detection
- **IEEE 802.11 allows:**
 - ❑ CSMA
 - ❑ CSMA/CA: reservations
 - ❑ polling from AP



Exposed Station Problem



C is exposed to transmission from A to B.

Use of handshaking in exposed station problem

